

SGA 7-I — Exposé VI

Source scan pages 71–96: pro-representability and the sheaves D^i

2. Pro-representable groupoids, continued

Proof of Proposition 2.14. We keep the notation of the preceding statement. Suppose that a homomorphism (σ_0, σ_1) of a normalized cogroupoid induces a $\mathcal{C}_\Lambda$ -equivalence on the associated cofibered groupoids. Applying the equivalence to the normalized base object gives a diagram of the form

$$\begin{array}{ccccc} B_{R_0}(x, e) & \longrightarrow & h_{R_0}(x, e) & \longrightarrow & h_{(R_0, R_1, \dots)}(x, e) \\ \downarrow & & \downarrow & & \downarrow h_{(\sigma_0, \sigma_1)} \\ B_{R_0}(x, e) & \longrightarrow & h_{R_0}(x, e) & \longrightarrow & h_{(R_0, R_1, \dots)}(x, e). \end{array}$$

Since, after passage to $k[\varepsilon]$, the relevant groupoid is totally disconnected, the induced tangent map is bijective. Hence the endomorphism $R_0 \rightarrow R_0$ is a surjective endomorphism of a complete local noetherian ring, and therefore an automorphism. The defining pull-back formula for R_1 then implies that $R_1 \rightarrow R_1$ is an automorphism as well. Thus (σ_0, σ_1) is an isomorphism.

Lemma 2.15. Let A, B be discrete groupoids over $\mathcal{C}_\Lambda$, and let $\varphi : A \rightarrow B$ be a minimally smooth functor. If A and B are homogeneous, then φ is a $\mathcal{C}_\Lambda$ -equivalence.

Proof. We must prove that $\varphi(S) : A(S) \rightarrow B(S)$ is bijective for every S . It is surjective by minimal smoothness, and bijective for $S = k[\varepsilon]$. We argue by induction on the length of S . If $S' \twoheadrightarrow S$ is a small extension, homogeneity gives

$$S' \times_S S' \simeq S' \times_k k[\text{Ker } f],$$

and therefore the commutative diagram

$$\begin{array}{ccc} A(S') \times_{A(S)} A(S') & \longrightarrow & A(S') \times A(k[\text{Ker } f]) \\ \varphi \times \varphi \downarrow & & \downarrow \varphi \times \varphi \\ B(S') \times_{B(S)} B(S') & \longrightarrow & B(S') \times B(k[\text{Ker } f]). \end{array}$$

Thus $A(S')$ and $B(S')$ are principal homogeneous sets under tangent groups identified by φ . If $\varphi(S)$ is bijective, then $\varphi(S')$ is injective and hence bijective. $\square$

Corollary 2.16. Let A be homogeneous over $\mathcal{C}_\Lambda$, with finite-dimensional tangent space. For every object ξ of A , the functor

$$\text{Isom}_\xi : \mathcal{C}_\Lambda / R \longrightarrow \text{Ens}, \quad R = P(\xi),$$

is pro-representable.

Proof. The functor Isom_ξ is homogeneous: for a diagram $T_1 \rightarrow T \leftarrow T_2$ with one arrow surjective,

$$\text{Isom}_\xi(T_1 \times_T T_2) \simeq \text{Isom}_\xi(T_1) \times_{\text{Isom}_\xi(T)} \text{Isom}_\xi(T_2).$$

The formal existence theorem gives a minimally smooth functor from a pro-representable functor to Isom_ξ ; Lemma 2.15 makes it an equivalence. $\square$

Theorem 2.17. A cofibered groupoid A over $\mathcal{C}_\Lambda$ is pro-representable by a normalized smooth cogroupoid if and only if

- (1) A is homogeneous;
- (2) $[A(k[\varepsilon])]$ and $\text{Aut}(1_{k[\varepsilon]})$ are finite-dimensional vector spaces over k .

Proof. Smooth cogroupoids are homogeneous, and the tangent and automorphism spaces are kernels and cokernels of maps between spaces $\text{Hom}(R_i, k[\varepsilon])$, hence finite-dimensional. Conversely, choose a minimally versal formal object ξ represented by R_0 . Corollary 2.16 pro-represents the functor of isomorphisms between the two pull-backs of ξ over $R_0 \hat{\otimes}_\Lambda R_0$ by an object R_1 , carrying a universal isomorphism. Composition of isomorphisms gives a cogroupoid $(R_0, R_1, \dots)$. The induced functor $h_{(R_0, R_1, \dots)} \rightarrow A$ is minimally smooth, hence, by Lemma 2.15, an equivalence. $\square$

Corollary 2.18. A functor $F : \mathcal{C}_\Lambda \rightarrow \text{Ens}$ is pro-representable if and only if it is homogeneous and $\dim_k F(k[\varepsilon]) < \infty$.

Corollary 2.19. Every smooth cogroupoid in $\hat{\mathcal{C}}_\Lambda$ can be normalized, up to isomorphism.

For a cogroupoid $(R_0, R_1, \dots)$, let $\bar{R}_1 = R_1/J$, where J is generated by $s_1(r) - s_2(r)$, $r \in R_0$. Then $(R_0, \bar{R}_1, \dots)$ defines a formal group scheme over R_0 .

Corollary 2.20. Let A be homogeneous and pro-represented by a normalized smooth cogroupoid $(R_0, R_1, \dots)$. Then $(R_0, \bar{R}_1, \dots)$ and $(k, k \hat{\otimes}_{R_0} \bar{R}_1, \dots)$ pro-represent the automorphism functors Aut_ξ and Aut^0 . Moreover the following are equivalent:

- (i) the formal group scheme is formally smooth over R_0 ;
- (ii) for every surjection $a' \rightarrow a$ in A , the map $\text{Aut}(a') \rightarrow \text{Aut}(a)$ is surjective;
- (iii) $\bar{R}_1$ pro-represents the functor of isomorphism classes over $\mathcal{C}_\Lambda$.

This is the combination of Proposition 2.7 with Lemma 2.15.

Remarks 2.21–2.22. Pulling a pro-representation back to $\mathcal{C}_\Lambda/k$ replaces the cogroupoid by its base change to k , preserving normalization and relative dimension. If A has two smooth pro-representations and one is normalized, then there is a comparison morphism of cogroupoids; the non-normalized base is a formal power-series extension of the normalized one, and the comparison becomes an isomorphism after this smooth base change.

3. The coherent sheaves D^i

We now recall the elementary obstruction theory of commutative algebra extensions in the language of Grothendieck cohomology on a site. Rings are commutative with unit. For an R -algebra A , let $\mathcal{C}_{A|R}$ be the category of arrows $B \rightarrow A$ in R -algebras, endowed with the topology whose coverings are surjective arrows. It has final object $A \rightarrow A$, initial object $R \rightarrow A$, fibre products and coproducts.

A sheaf F of A -modules on $\mathcal{C}_{A|R}$ is a functor such that, for every cover $B' \twoheadrightarrow B$,

$$0 \rightarrow F(B) \rightarrow F(B') \rightrightarrows F(B' \times_B B')$$

is exact. Additive sheaves preserve coproducts. An A -module M gives the additive sheaf

$$\text{Der}_R(-, M) : B \longmapsto \text{Der}_R(B, M),$$

where M is viewed as a B -module through $B \rightarrow A$.

Let $\tilde{\mathcal{C}}_{A|R}$ and $\hat{\mathcal{C}}_{A|R}$ denote sheaves and presheaves. The inclusion of sheaves into presheaves has right derived functors $\mathcal{H}_{A|R}^q$, and

$$H^q(A|R, F) = [\mathcal{H}_{A|R}^q(F)](A).$$

An object $B \in \mathcal{C}_{A|R}$ and a ring homomorphism $S \rightarrow R$ give the functorial diagrams

$$\begin{array}{ccc} \widehat{\mathcal{C}}_{B|R} & \longrightarrow & \widehat{\mathcal{C}}_{A|R} \\ \uparrow & & \uparrow \\ \widetilde{\mathcal{C}}_{B|R} & \longrightarrow & \widetilde{\mathcal{C}}_{A|R} \end{array} \quad \begin{array}{ccc} \widehat{\mathcal{C}}_{A|S} & \longrightarrow & \widehat{\mathcal{C}}_{A|R} \\ \uparrow & & \uparrow \\ \widetilde{\mathcal{C}}_{A|S} & \longrightarrow & \widetilde{\mathcal{C}}_{A|R}. \end{array}$$

For a covering $B' \rightarrow B$, the groups $H^q(\{B' \rightarrow B\}, F)$ are the cohomology groups of the Čech semi-simplicial module

$$F(B') \rightrightarrows F(B' \times_B B') \rightrightarrows F(B' \times_B B' \times_B B') \rightrightarrows \cdots.$$

For a presheaf F , set

$$\mathcal{H}_{A|R}^q(F)(B) = \varinjlim_{\{B' \rightarrow B\}} H^q(\{B' \rightarrow B\}, F).$$

There is the usual spectral sequence

$$H^p(A|R, \mathcal{H}^q(F)) \Rightarrow H^{p+q}(A|R, F),$$

and hence, in low degree,

$$\begin{aligned} H^1(A|R, F) &= \mathcal{H}^1(A|R, F), \\ 0 \rightarrow \mathcal{H}^2(A|R, F) &\rightarrow H^2(A|R, F) \rightarrow H^1(A|R, \mathcal{H}^1(F)) \\ &\rightarrow \mathcal{H}^3(A|R, F). \end{aligned}$$

Lemma 3.2. If $P \twoheadrightarrow B$ is a cover by a polynomial R -algebra, then

$$H^n(\{P \rightarrow B\}, F) \xrightarrow{\sim} \mathcal{H}_{A|R}^n(F)(B).$$

Polynomial covers dominate all coverings, which proves the lemma.

Corollary 3.3. (a) If A is noetherian and F is of finite type, then $\mathcal{H}^n(F)$ is of finite type. (b) If P is polynomial over R , then $H^n(P|R, F) = 0$ for all $n > 0$. (c) Exactness of sheaves may be checked on polynomial R -algebras. Indeed, the polynomial subsite is cofinal and every surjection in it admits a section.

Lemma 3.5. For a diagram of R -algebras

$$\begin{array}{ccc} B' & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & A \end{array}$$

and an R -algebra S :

(i) if $\mathrm{Tor}_1^R(B, S) = 0$, then

$$(B' \times_B C) \otimes_R S \simeq (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S);$$

(ii) if B' is R -flat and $\mathrm{Tor}_i^R(B, S) = \mathrm{Tor}_i^R(C, S) = 0$ for $0 < i \leq n$, then $\mathrm{Tor}_i^R(B' \times_B C, S) = 0$ for $0 < i \leq n$.

The proof compares the exact rows obtained from $0 \rightarrow J \rightarrow B' \rightarrow B \rightarrow 0$:

$$\begin{array}{ccccccc} 0 & \longrightarrow & J \otimes_R S & \longrightarrow & (B' \times_B C) \otimes_R S & \longrightarrow & C \otimes_R S \longrightarrow 0 \\ & & \parallel & & \downarrow & & \parallel \\ 0 & \longrightarrow & J \otimes_R S & \longrightarrow & (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S) & \longrightarrow & C \otimes_R S \longrightarrow 0. \end{array}$$

Corollary 3.6. If $\mathrm{Tor}_1^R(A, S) = 0$, then

$$H^n(A \otimes_R S|S, G) \simeq H^n(A|R, SG).$$

If S is R -flat, this gives $S\mathcal{H}^n(G) \simeq \mathcal{H}^n(SG)$. Polynomial covers remain Tor-acyclic under the hypotheses of Lemma 3.5.

Proposition 3.7. (a) If $\mathrm{Tor}_i^R(A, S) = 0$ for $0 < i \leq n$, then

$$\mathcal{H}^i(A \otimes_R S|S, F) \xrightarrow{\sim} \mathcal{H}^i(A|R, SF), \quad i \leq n.$$

(b) If P is polynomial over R , then for an additive sheaf F on $\mathcal{C}_{A \otimes_R P|R}$,

$$H^n(A \otimes_R P|R, F) \simeq H^n(A|R, F), \quad n > 0.$$

(c) For $S \rightarrow R \rightarrow A$ and additive F on $\mathcal{C}_{A|S}$, there is an exact sequence

$$0 \rightarrow H^0(A|R, F_R) \rightarrow H^0(A|S, F) \rightarrow H^0(R|S, F) \rightarrow H^1(A|R, F_R) \rightarrow \cdots,$$

where $F_R(X) = \mathrm{Ker}(F(X) \rightarrow F(R))$. The proof compares the spectral sequences

$$\begin{array}{ccc} H^p(A \otimes_R S|S, \mathcal{H}^q(F)) & \Longrightarrow & H^{p+q}(A \otimes_R S|S, F) \\ \downarrow & & \downarrow \\ H^p(A|R, \mathcal{H}^q(SF)) & \Longrightarrow & H^{p+q}(A|R, SF) \end{array}$$

and then uses the Leray sequence for the forgetful functor.

Corollary 3.8. If $\mathrm{Tor}_i^R(A, B) = 0$ for $0 < i \leq n$, then for additive F

$$H^i(A \otimes_R B|R, F) \simeq H^i(A|R, F) \oplus H^i(B|R, F), \quad i \leq n.$$

This follows from the splitting diagram

$$\begin{array}{ccc} H^i(A \otimes_R B|A, F_A) & \longrightarrow & H^i(A \otimes_R B|R, F) \\ \sim \downarrow & & \\ H^i(B|R, F_A) & \longrightarrow & H^i(B|R, F). \end{array}$$

If moreover $A \otimes_R A \simeq A$ and $\mathrm{Tor}_i^R(A, A) = 0$, then $H^i(A|R, F) = 0$ for $i > 0$.

Corollary 3.9. For a multiplicative subset $S \subset A$, if F and F_S are additive, then

$$H^i(S^{-1}A|R, F) \simeq H^i(A|R, F).$$

For an A -module M , define

$$D^i(A|R, M) = H^i(A|R, \mathrm{Der}_R(-, M)).$$

Choose $P \twoheadrightarrow A$ polynomial over R , with kernel I .

Lemma 3.10.

$$\begin{aligned} D^1(A|R, M) &= \text{Coker}(\text{Der}_R(P, M) \rightarrow \text{Hom}_A(I/I^2, M)), \\ D^2(A|R, M) &= H^1(A|R, \mathcal{H}^1(\text{Der}_R(-, M))). \end{aligned}$$

The proof uses the simplicial algebra $P^{(n)} = P \times_A \cdots \times_A P$ and the exact sequence

$$0 \rightarrow \text{Der}_R(A, M) \rightarrow \text{Der}_R(P, M) \rightarrow \text{Hom}_A(I/I^2, M) \rightarrow H^1(\{P \rightarrow A\}, \text{Der}_R(-, M)) \rightarrow 0.$$

Let $F \twoheadrightarrow I$ be a free P -module mapping onto I , and let $K_\bullet$ be the associated Koszul complex.

Lemma 3.11.

$$D^2(A|R, M) = \text{Coker}(H^1(K_\bullet, M) \rightarrow \text{Hom}_A(H_1(K_\bullet), M)).$$

This follows by replacing the cover $P \rightarrow A$ by the polynomial algebra $P[F]$; the obstruction term is

$$(F) \cap (F') / ((F)(F')) \simeq \text{Tor}_1^P(A, A) = H_1(K_\bullet).$$

Theorem 3.12. The groups D^i satisfy:

- (1) a long exact sequence in $D^i(A|R, -)$ for every exact sequence of A -modules;
- (2) for $S \rightarrow R \rightarrow A$, an exact sequence

$$0 \rightarrow D^0(A|R, M) \rightarrow D^0(A|S, M) \rightarrow D^0(R|S, M) \rightarrow D^1(A|R, M) \rightarrow \cdots;$$

- (3) Tor-independent base change $D^i(A \otimes_R S|S, M) \simeq D^i(A|R, M)$;
- (4) localization $D^i(S^{-1}A|R, S^{-1}M) \simeq S^{-1}D^i(A|R, M)$;
- (5) the product formula $D^i(A \otimes_R B|R, M) \simeq D^i(A|R, M) \oplus D^i(B|R, M)$ under Tor-independence;
- (6) compatibility with tensoring by a flat A -module, when A is essentially of finite type over a noetherian R ;
- (7) finiteness of $D^i(A|R, M)$ for M finite;
- (8) the explicit formulas

$$\begin{aligned} D^0(A|R, M) &= \text{Der}_R(A, M), \\ D^1(A|R, M) &= \text{Coker}(\text{Der}_R(P, M) \rightarrow \text{Hom}_A(I/I^2, M)) = \text{Exalcomm}(A|R, M), \\ D^2(A|R, M) &= \text{Coker}(H^1(K_\bullet, M) \rightarrow \text{Hom}_A(H_1(K_\bullet), M)). \end{aligned}$$

The extension interpretation of D^1 is given by the push-out diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & I & \longrightarrow & P & \longrightarrow & A \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & M & \longrightarrow & A' & \longrightarrow & A \longrightarrow 0. \end{array}$$

Two maps $I \rightarrow M$ differing by a derivation of P give isomorphic extensions, and every extension arises this way.

Corollary 3.13.

- (a) A is formally smooth over R iff $D^1(A|R, M) = 0$ for every M .
- (b) If R is noetherian and A finite type over R , then A is a relative complete intersection iff $D^2(A|R, M) = 0$ for every M .
- (c) Localization commutes with D^i .
- (d) If A is local, essentially of finite type over noetherian R , and $\hat{A}$ is its completion, then

$$D^i(\hat{A}|R, \hat{A} \otimes_A E) \simeq \hat{A} \otimes_A D^i(A|R, E)$$

for every finite A -module E .

For (d), the final step compares algebra extensions and uses Artin–Rees. The completion comparison is expressed by the exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \hat{E} & \longrightarrow & B & \longrightarrow & A & \longrightarrow & 0 \\ & & \parallel & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \hat{E} & \longrightarrow & \hat{B} & \longrightarrow & \hat{A} & \longrightarrow & 0, \end{array}$$

which proves the needed injectivity and surjectivity for D^1 , and hence the assertion.